RestNet

UNDERSTAND YOUR SLEEP

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NAVIGATION

- Sleep Dashboard
- Night Explorer
- My Sleep Insights

Environmental Sensing for Sleep Quality Analysis

IoT & Applications
Imperial College London
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My Sleep Insights

What Helps You Sleep Best?

Compare your **best and worst nights**, explore how **bedroom conditions correlated with sleep**, see your **optimal ranges** on good vs poor nights, and get an **actionable takeaway**.

Best vs Worst Nights

Your highest and lowest scoring nights side by side. Same bed, different conditions.

BEST NIGHT: THURSDAY, 19 FEBRUARY

96 / 100 SLEEP SCORE

7h 44m sleep · 23% deep · 33% REM · 35% light · 6% awake

BEDROOM CONDITIONS

Average temp: 18.5°C (range: 1.5°C)
Average humidity: 43%
Average noise: 76 (variability: 19.6)

WORST NIGHT: SATURDAY, 14 FEBRUARY

66 / 100 SLEEP SCORE

7h 32m sleep · 28% deep · 48% REM · 13% light · 17% awake

BEDROOM CONDITIONS

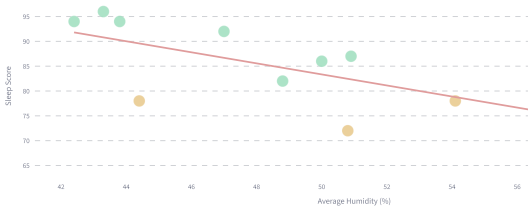
Average temp: 20.5°C (range: 1.5°C)
Average humidity: 58%
Average noise: 76 (variability: 22.7)

Correlation Explorer

How each bedroom condition correlated with your sleep score across 14 nights.

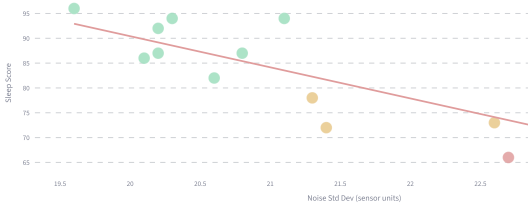
● Good sleep (score ≥ 80) ● Fair sleep (70–79) ● Poor sleep (score < 70)

Humidity
Spearman r = -0.70
Higher humidity linked to lower sleep scores.

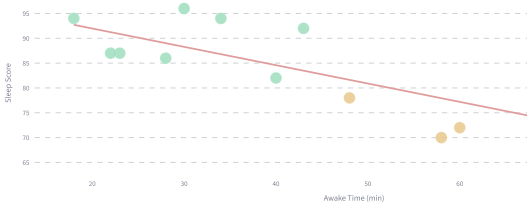


Noise Level
Spearman r = -0.69
Noisier nights meant worse sleep.

Noise Variability
Spearman r = -0.79
Inconsistent noise was the strongest predictor of poor sleep.



Awake Time
Spearman r = -0.77
More time awake during the night correlated with lower scores.



Temperature, PM2.5, and NO_2 were also tested but showed no significant correlation with sleep quality.

Your Optimal Ranges

Median-split comparison: 7 good nights (score ≥ 84) vs 7 poor nights (score < 84). Only the four significant predictors are shown.

Condition	Good Nights Avg	Poor Nights Avg	Difference
Noise Variability ⭐	20.3	22.2	1.9 steadier
Humidity	48.1%	53.4%	5.3% drier
Noise Level	70.7	86.0	15.2 quieter
Temp Range	1.5°C	1.6°C	0.1°C more stable

Actionable Takeaway

What you can do to improve your sleep, based on 14 nights of data.

🔊 Why noise variability matters

It's not about how loud your room is overall. It's the **sudden jumps in noise** (traffic, doors, neighbours) that pull you out of deep sleep without fully waking you up. The WHO Night Noise Guidelines found that **intermittent noise is more disruptive than steady noise** at the same volume, because your brain keeps reacting to each new sound even while you sleep.

On your **best nights**, noise variability averaged 20.3. On your **worst nights**, it jumped to 22.2. That difference was associated with a gap of about 17 points in sleep score.

🎯 Things worth trying

- **White noise machine or fan** to create a steady background hum that masks sudden spikes
- **Earplugs (NRR 20+)** for a low-tech fix on noisy nights
- **Close windows** facing busy roads, especially late evening when traffic noise tends to peak
- **Move your bed** away from the noisiest wall if possible

The goal isn't silence. A steady, predictable sound is better than **quiet that gets interrupted by random spikes.**

These insights are based on 14 nights. As your dataset grows, recommendations will become more reliable.